

Vignesh S | Python & Data Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Versatile Computer Science post-graduate with over 1+ year of professional experience bridging Python backend software development and cloud data engineering. Proficient in building secure REST APIs using Flask, automating enterprise data pipelines using PySpark and Azure, and managing relational databases. Experienced in resolving complex system bottlenecks and infrastructure support tickets to ensure high application availability.

Technical Skills

Programming Languages:

Python, SQL, Javascript

Backend Development:

Flask, REST APIs, JWT Authentication, CI/CD Pipelines, SQLAlchemy, Pandas

Database & Storage:

PostgreSQL, SQL Server, Azure Data Lake Gen2

Data Engineering:

Azure Data Factory (ADF), Azure Databricks, Medallion Architecture, ETL/ELT

Tools & Technologies:

Git, GitHub, Docker, Postman, Power BI, Service Desk/Ticketing Systems, Manage Engine

Operating Systems:

Linux, Windows

Cloud:

Microsoft Azure (AZ-900), AWS Cloud Practitioner

Frontend:

HTML, CSS, JS, React (Basic)

Professional Experience

SQL Developer | Balbhas Business Sysnomics LLC

January 2026 – Present

- Engineered optimized SQL queries and complex stored procedures in MS SQL Server to automate enterprise data extraction, transformation, and reporting.
- Managed cloud support ticketing queues by diagnosing and troubleshooting production data quality issues under strict SLAs.
- Investigated and resolved database bottlenecks, significantly enhancing reporting efficiency, internal ERP application workflows, and overall system performance.
- Collaborated with application development teams to restructure database schemas and align backend operational databases with business needs.
- Generated operational and business reports using custom SQL-based analytical solutions.

Web & Database Developer | Cloud Hawk

January 2024 – February 2025

- Designed and maintained Python/Flask web applications, ensuring responsive backend data retrieval and processing workflows for commercial clients.
- Utilized Pandas and NumPy to perform data cleaning, array manipulations, and structural file formatting before loading data into relational systems.
- Administered frontend visual customizations utilizing HTML5, CSS3, and JavaScript to interface seamlessly with REST endpoints.

- Owned deployment support pipelines, troubleshooting application defects and deploying functional patches across client staging environments via Git.

Projects

Workforce Management System

Tech Stack: Python, Flask, PostgreSQL, JWT, Docker, Postman

- Developed a secure backend automation platform featuring robust role-based access control (RBAC) to monitor digital timesheets and ERP workflows.
- Secured critical application endpoints by implementing stateless JWT user authentication and authorization logic.
- Designed normalized relational PostgreSQL schemas and abstracted database queries using SQLAlchemy ORM.
- Containerized the microservice infrastructure using Docker to guarantee cross-environment deployment reliability.

Doctor Appointment Management System

Tech Stack: Python, Flask, PostgreSQL, Docker, Git

- Developed a web application for managing doctor appointments and patient records.
- Implemented authentication, CRUD operations, and database integration.
- Designed and managed PostgreSQL database tables and relationships.
- Utilized Docker and Git-based workflows for development and deployment.

Azure Data Engineering Pipeline

Tech Stack: Python, Pandas, NumPy, Azure Data Factory, Azure Databricks, PySpark, ADLS Gen2, Power BI

- Architected an end-to-end cloud data pipeline automating the ingestion, cleaning, and transformation of enterprise sales datasets.
- Leveraged Pandas and NumPy within Databricks notebooks to execute initial data transformations, numerical computations, and outlier detection.
- Implemented Medallion Architecture (Bronze, Silver, Gold stages) within Azure Data Lake to establish data compliance standards.
- Orchestrated ETL/ELT workflows via Azure Data Factory and scaled heavy data mutations using PySpark in Databricks.
- Built interactive Power BI dashboards to map critical regional operational metrics and sales KPIs for stakeholders.

Certifications

Microsoft Azure Fundamentals (AZ-900)

AWS Certified Cloud Practitioner (CCP)

Python & SQL

Education

Bachelor's Degree in Computer Science
Kongunadu Arts & Science College
2020

Master's Degree in Computer Science
Scott Christian College of Arts & Science
2022